

Pathophysiology: Acid-Base Disorders

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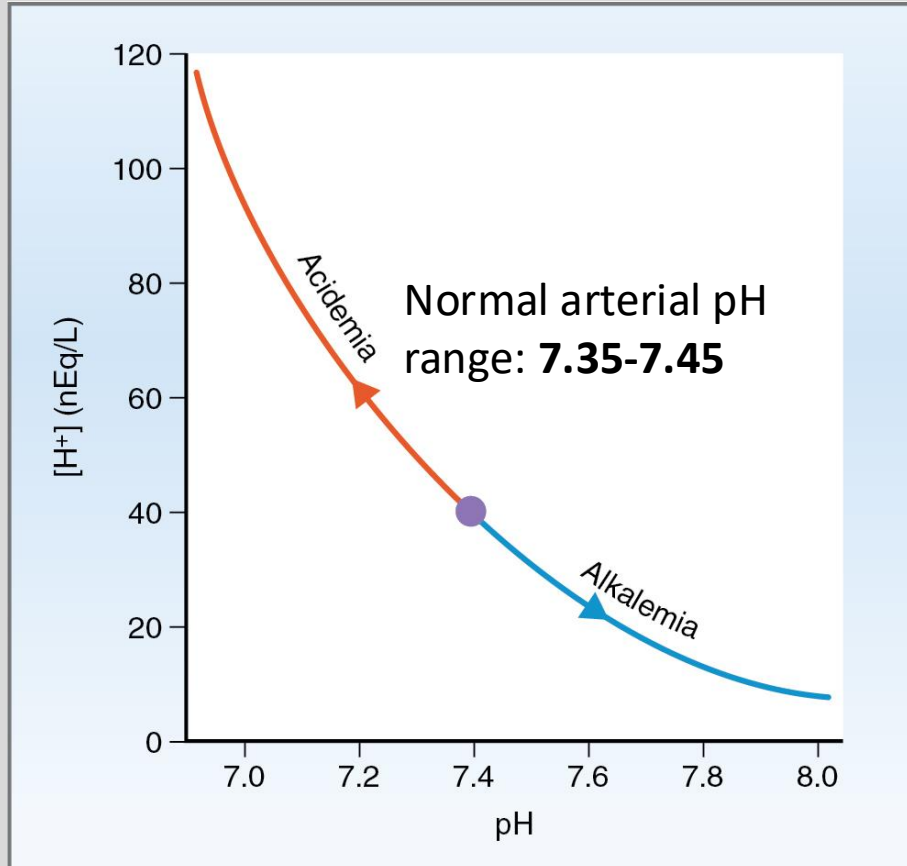


Do.
Make.
Innovate.
Reinvent the Future.

Objectives:

- Describe the kinetics of weak acids and bases and how the Henderson-Hasselbalch Equation can be useful in the determination of acid-base disorders.
- Describe the utility of an acid-base map for the categorization of simple or mixed acid-base disorders.
- Describe the utility of the anion gap of plasma in the diagnosis of acid-base disorders and calculate and interpret examples.
- Compare and contrast the four major types of simple acid-base disorders in their causes and the body's regulatory mechanisms to regain homeostasis.
- Describe the sequence of events for each of the four major types of simple acid-base disorders.
- Describe and apply a diagnostic workflow through the four major types of simple acid-base disorders. Use the 'renal rules' for calculations on hidden acid-base disorders.

Non-Linear Alkaline vs. Acidic pH Changes



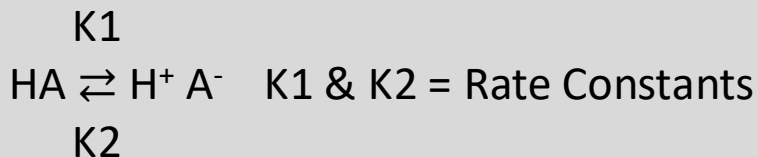
$$pH = -\log_{10}[H^+]$$

$$\text{Arterial } H^+ = [40 \times 10^{-9} \text{ Eq/L}]$$

$$pH = -\log_{10} [40 \times 10^{-9} \text{ Eq/L}] = 7.4$$

$$\text{Normal pH in ICF} = 7.2$$

Kinetics of Weak Acids & Bases



Combining Rate Constants: $K = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$

K = Equilibrium Constant

Rearrange: $[\text{H}^+] = K \frac{[\text{HA}]}{[\text{A}^-]}$

Express as pH: $-\log[\text{H}^+] = -\log K - \log \frac{[\text{HA}]}{[\text{A}^-]}$

$$\text{pH} = \text{pKa} + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Henderson-Hasselbalch Equation

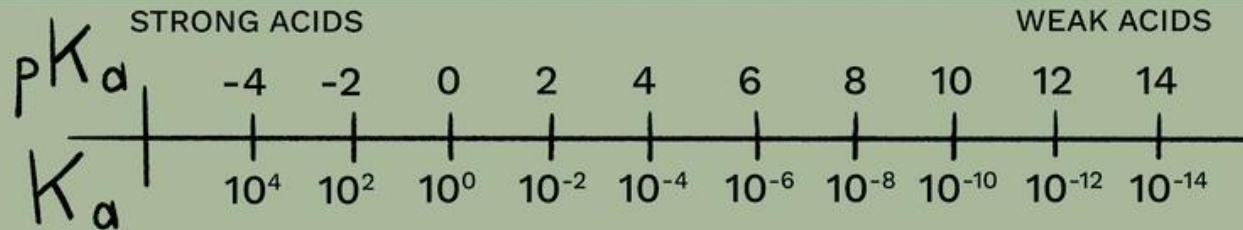
Strong and Weak Acids

What Is pK_a in Chemistry?

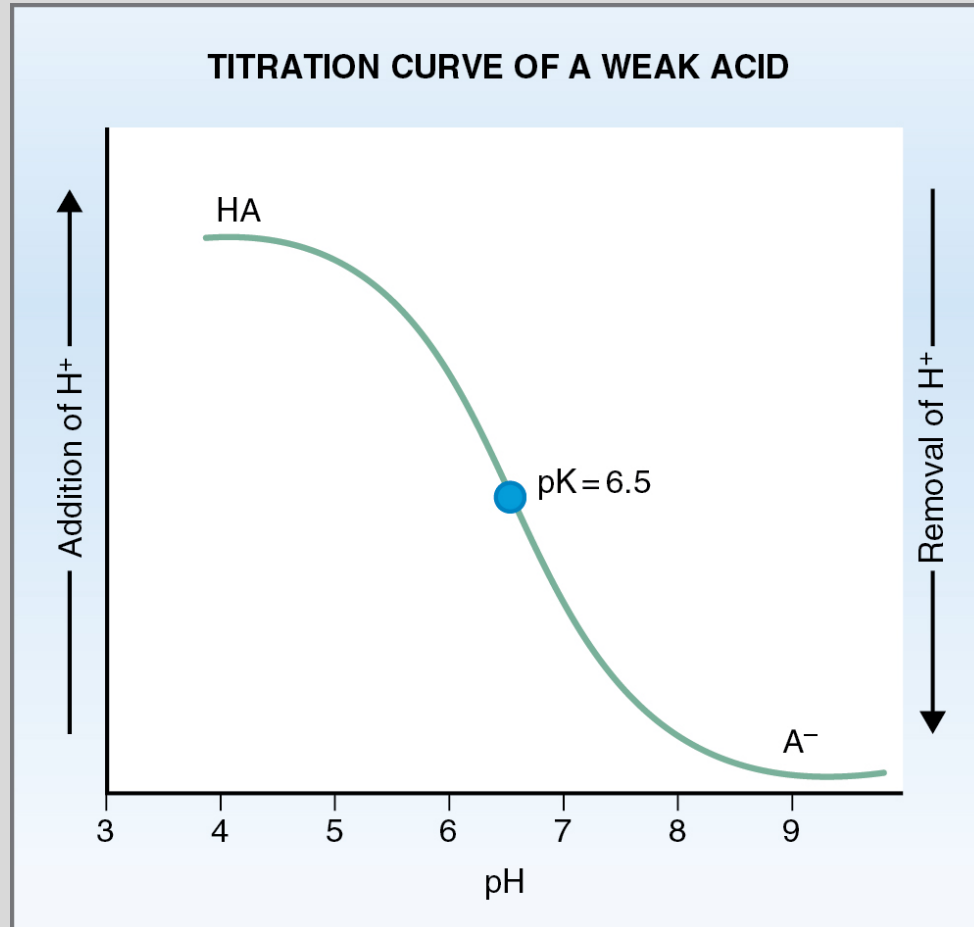
The pK_a value is the negative base-10 logarithm of the acid dissociation constant (K_a) of a solution.

$$pK_a = -\log_{10} K_a$$

The smaller the value of pK_a , the stronger the acid.



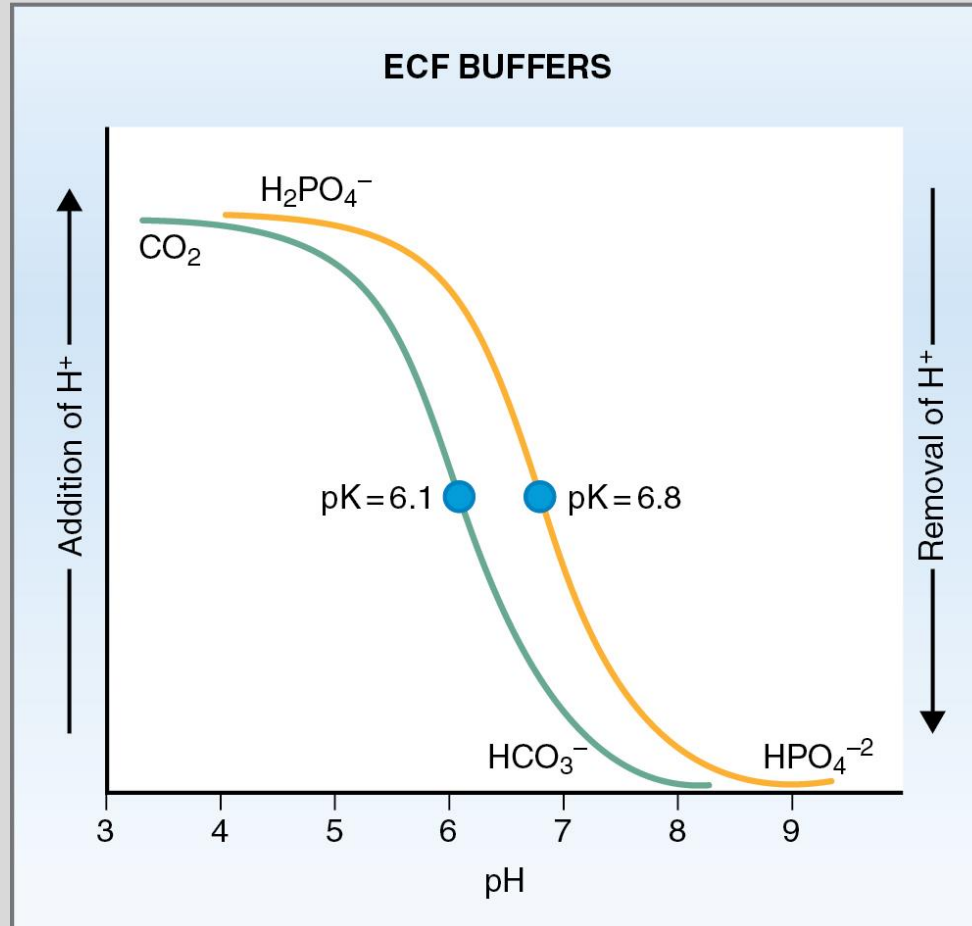
At the pKa $[HA]$ and $[A^-]$ are Equal



Linear portion of sigmoidal curve:
Most effective buffering

Bicarbonate & Phosphate

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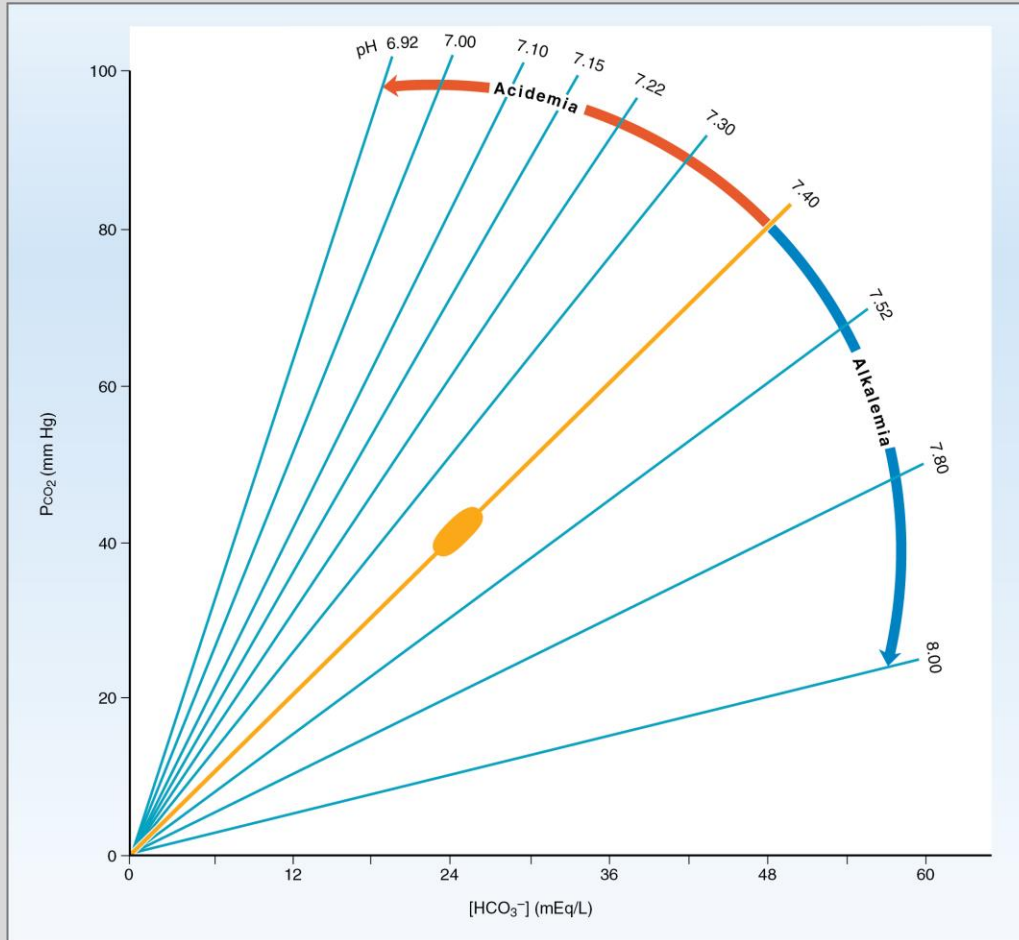


HA Form

A⁻ Form

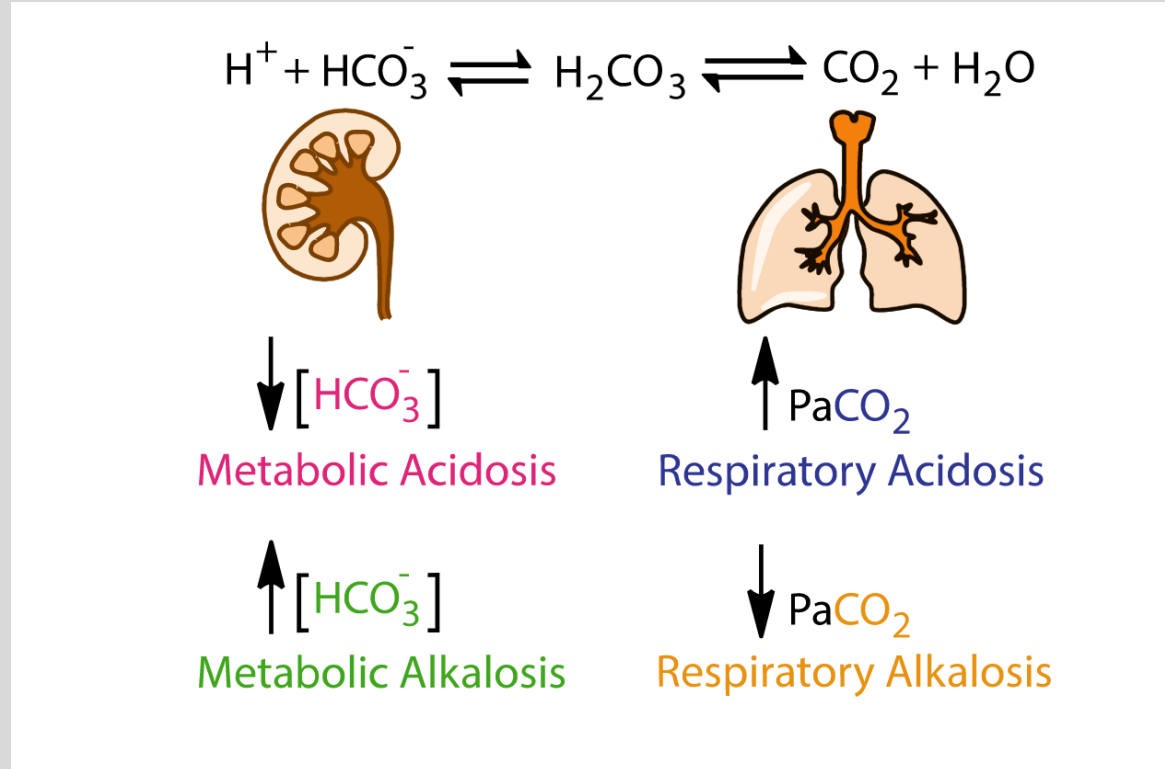
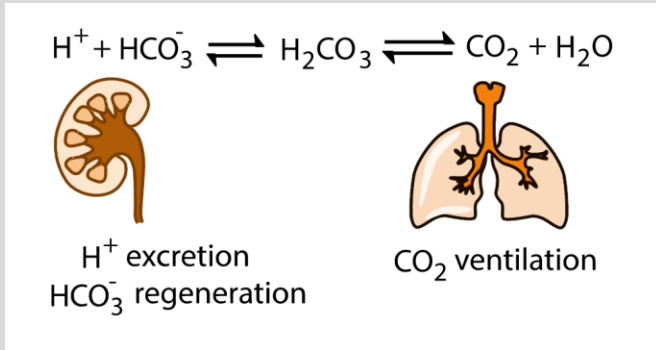
Acid-Base-Map: PCO_2 , $[\text{HCO}_3^-]$, & pH

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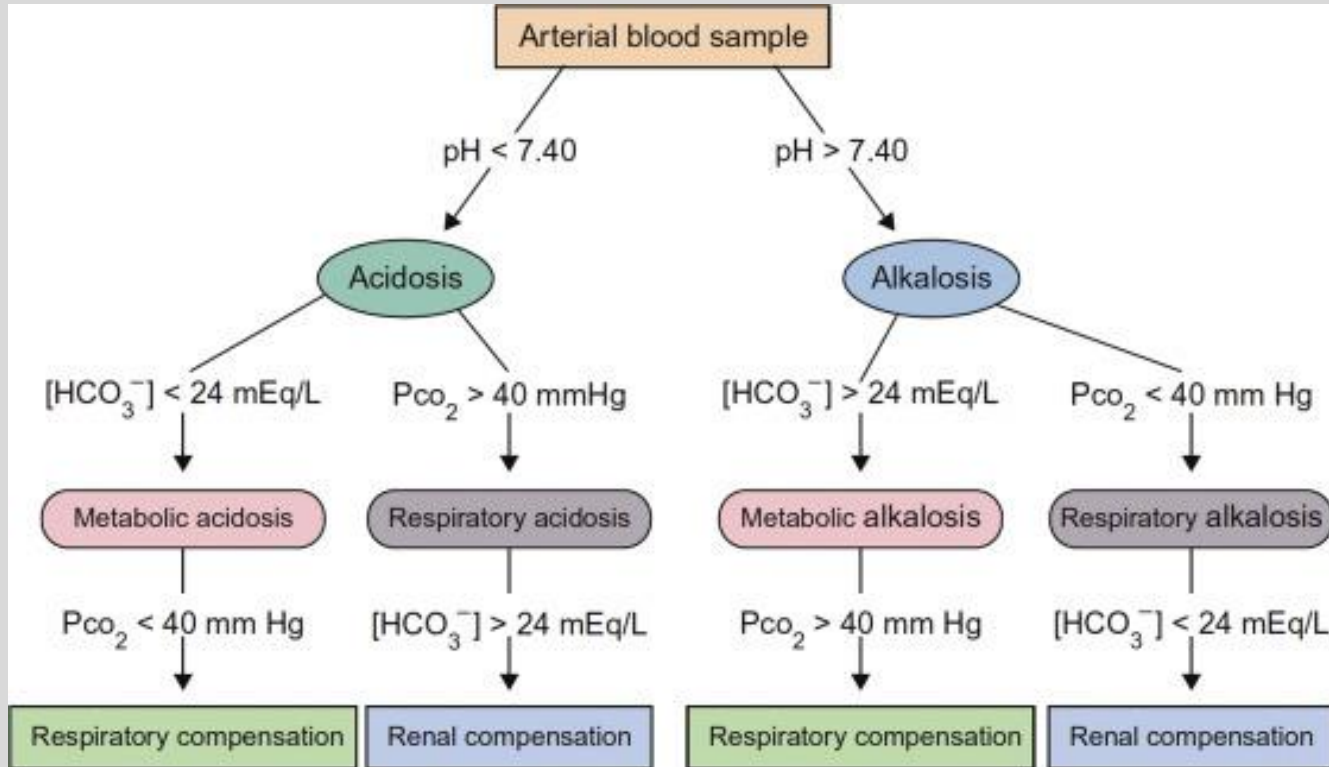
Various combinations can yield
pH 7.4 → **Isohydric lines**

Metabolic vs. Respiratory Acid-Base Disturbances



Four Simple Acid-Base Disorders

More at One Time = Mixed



**Respiratory
compensation followed
by renal compensation!**

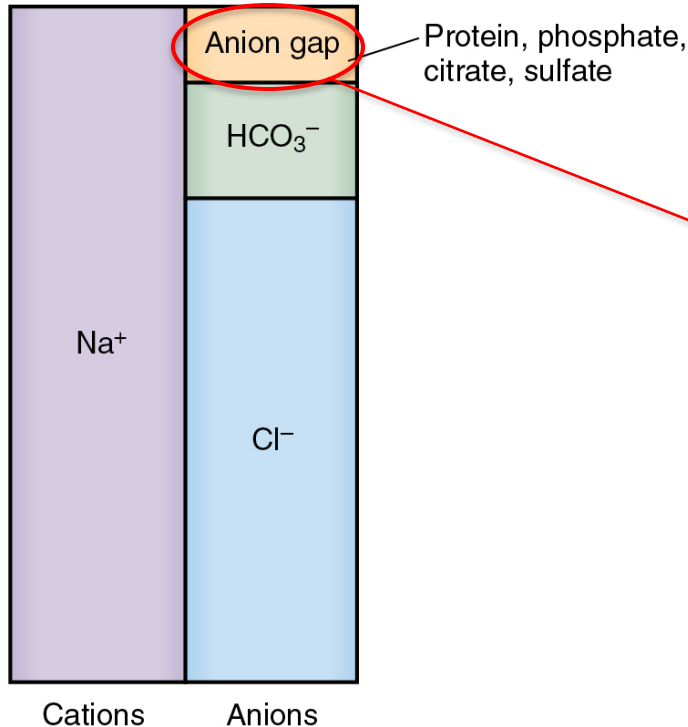
Summary of Simple Acid-Base Disorders

Disorder	$\text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{HCO}_3^-$	Respiratory Compensation	Renal Compensation or Correction
Metabolic Acidosis	↓	Hyperventilation	↑ HCO_3^- reabsorption (correction)
Metabolic Alkalosis	↑	Hypoventilation	↑ HCO_3^- excretion (correction)
Respiratory Acidosis	↑	None	↑ HCO_3^- reabsorption (compensation)
Respiratory Alkalosis	↓	None	↓ HCO_3^- reabsorption (compensation)

Bold arrows indicate initial disturbance.

Diagnosis of Acid-Base Disorders & the Concept of Electroneutrality

ANION GAP OF PLASMA



Unmeasured anions

Useful in differential diagnosis
of **metabolic acidosis**

Calculating the Plasma Anion Gap

Normal Range: 8–14 mEq/L

Plasma anion gap = $[\text{Na}^+] - ([\text{HCO}_3^-] + [\text{Cl}^-])$

Where:

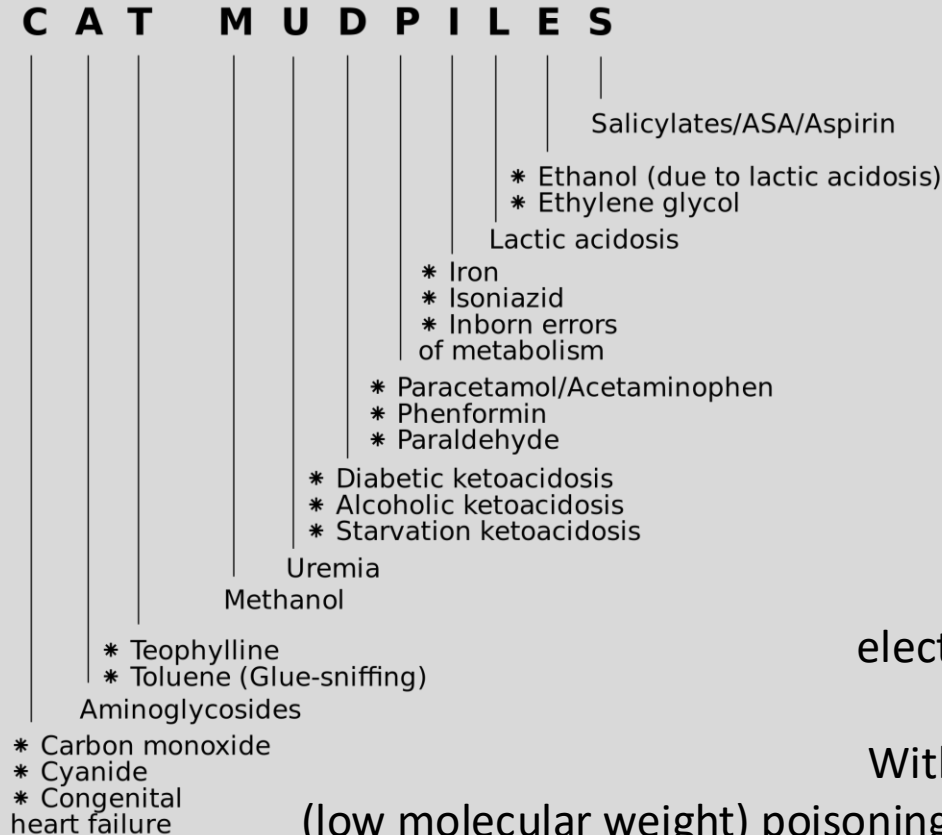
Plasma anion gap = Unmeasured anions (mEq/L)

$[\text{Na}^+]$ = Measured cation (mEq/L)

$[\text{HCO}_3^-]$ and $[\text{Cl}^-]$ = Measured anions (mEq/L)

Thus, if the **Na^+** concentration is **140 mEq/L**, the **HCO_3^-** concentration is **24 mEq/L**, and the **Cl^-** concentration is **104 mEq/L**, then the plasma anion gap is **12 mEq/L**.

Causes for Metabolic Acidosis with an Increased Anion Gap



To preserve electroneutrality HCO_3^- is reduced

With methanol or ethylene glycol (low molecular weight) poisoning there can be an osmolar gap

A Second (Hidden) Metabolic Acid-Base Disorder may Change the Blood HCO_3^-

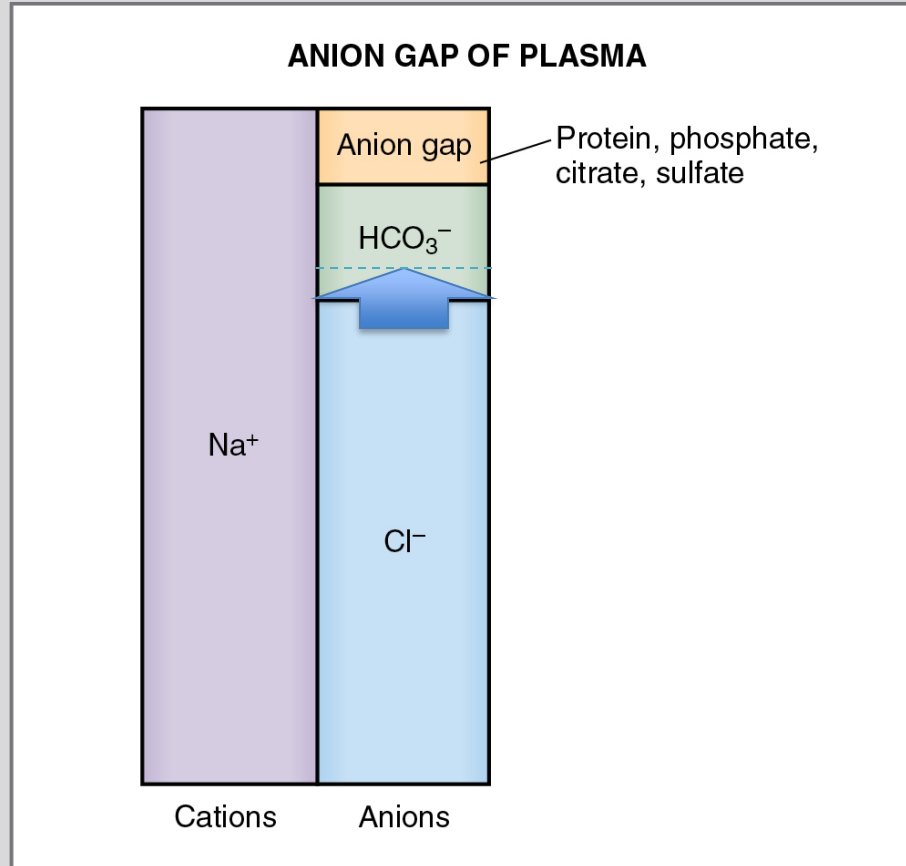
Δ/Δ tool ($\Delta \text{HCO}_3^- / \Delta$ anion gap)

If **only** metabolic acidosis with increased anion gap $\rightarrow$ **decreased blood HCO_3^- (from normal) = increase in anion gap**

If a second metabolic disorder is **increasing blood HCO_3^-** (i.e., metabolic alkalosis) $\rightarrow$ **decrease blood $\text{HCO}_3^- < \text{increase in anion gap}$**

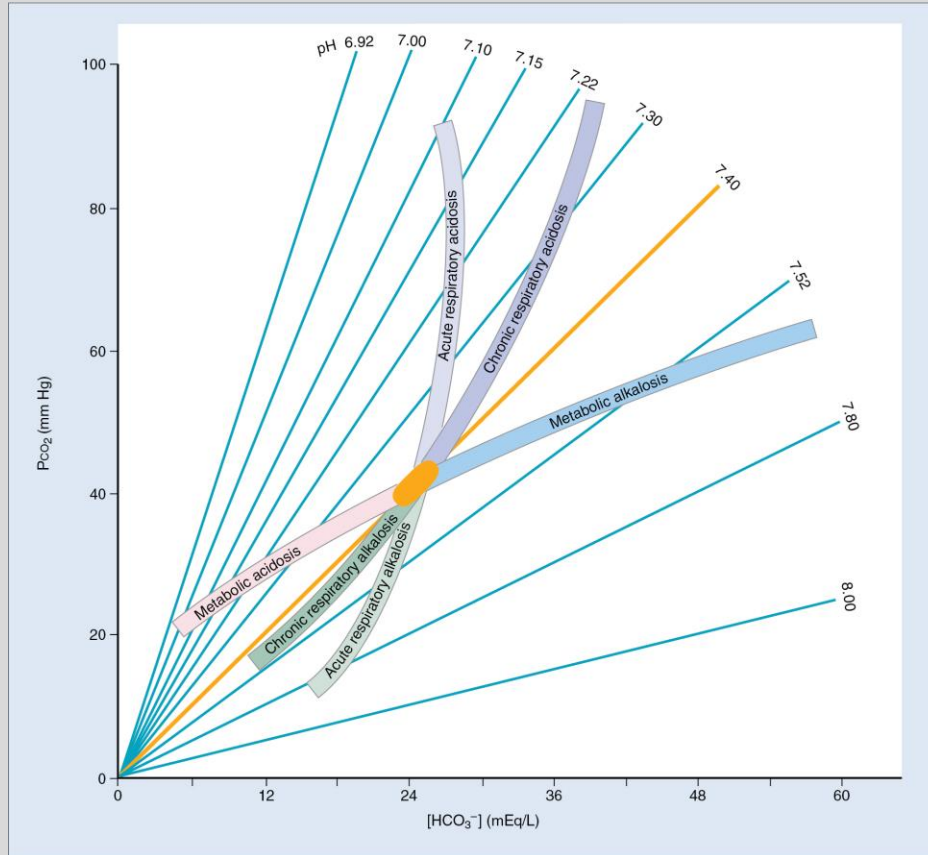
If a second metabolic disorder is **decreasing blood HCO_3^-** (i.e., metabolic acidosis with normal anion gap) $\rightarrow$ **decrease blood $\text{HCO}_3^- > \text{increase in anion gap}$**

Hyperchloremic Metabolic Acidosis with a Normal Anion Gap



Acid Base Map

Assessing a patient's acid-base status



Metabolic Disorders:
Compensation occurs immediately

Respiratory Disorders:
Compensation occurs over days

If patient's values fall in between 2
shaded areas → Mixed disorder

Renal Rules for Predicting Compensatory Responses in Simple Acid-Base Disorders

Acid-Base Disturbance	Primary Disturbance	Compensation	Predicted Compensatory Response
Metabolic Acidosis	$\downarrow [\text{HCO}_3^-]$	$\downarrow \text{Pco}_2$	1 mEq/L decrease in $\text{HCO}_3^- \rightarrow 1.2 \text{ mm Hg decrease in } \text{Pco}_2$
Metabolic Alkalosis	$\uparrow [\text{HCO}_3^-]$	$\uparrow \text{Pco}_2$	1 mEq/L increase in $\text{HCO}_3^- \rightarrow 0.6 \text{ mm Hg increase in } \text{Pco}_2$
Respiratory Acidosis			
Acute	$\uparrow \text{Pco}_2$	$\uparrow [\text{HCO}_3^-]$	1 mm Hg increase in $\text{Pco}_2 \rightarrow 0.1 \text{ mEq/L increase in } \text{HCO}_3^-$
Chronic	$\uparrow \text{Pco}_2$	$\uparrow [\text{HCO}_3^-]$	1 mm Hg increase in $\text{Pco}_2 \rightarrow 0.3 \text{ mEq/L increase in } \text{HCO}_3^-$
Respiratory Alkalosis			
Acute	$\downarrow \text{Pco}_2$	$\downarrow [\text{HCO}_3^-]$	1 mm Hg decrease in $\text{Pco}_2 \rightarrow 0.2 \text{ mEq/L decrease in } \text{HCO}_3^-$
Chronic	$\downarrow \text{Pco}_2$	$\downarrow [\text{HCO}_3^-]$	1 mm Hg decrease in $\text{Pco}_2 \rightarrow 0.4 \text{ mEq/L decrease in } \text{HCO}_3^-$

Metabolic Alkalosis – Increased HCO_3^-

Arterial blood profile: **Increased pH, increased HCO_3^- and increased PCO_2**

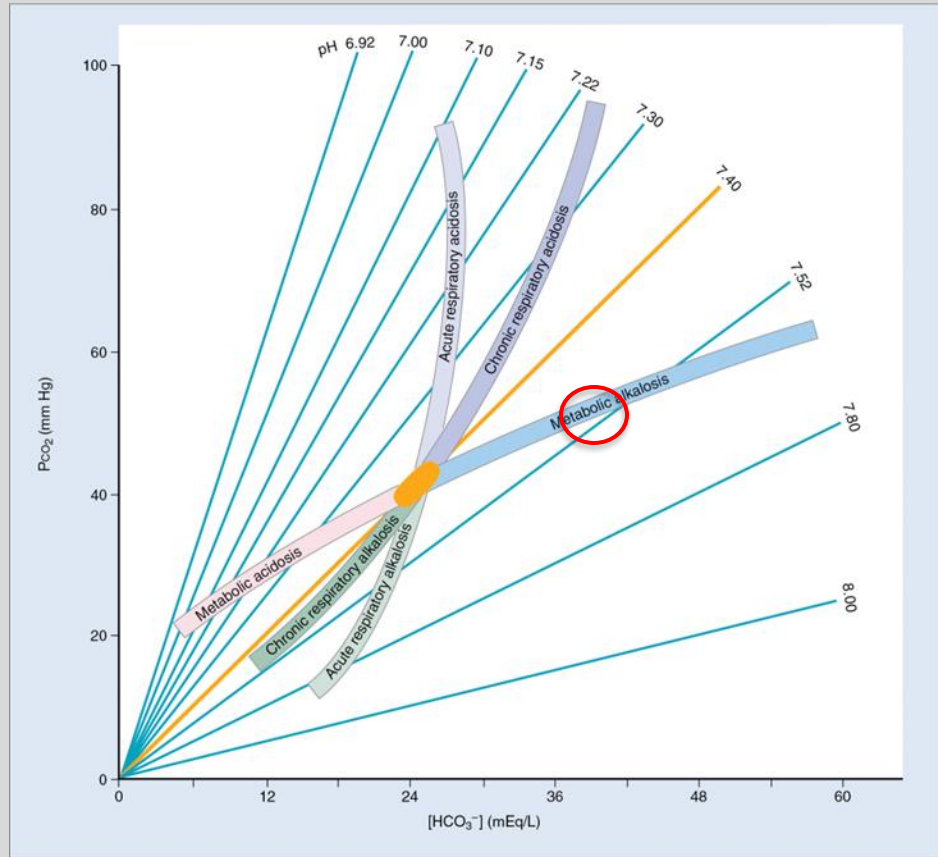
CAUSES

- **Excessive loss of fixed H^+**
Vomiting, hyperaldosteronism
- **Gain of HCO_3^-**
Ingestion of NaHCO_3 (e.g., heartburn medication), Milk-alkali syndrome
- **Volume contraction alkalosis**
Loop or thiazide diuretics → increased reabsorption of HCO_3^-

Sample Case #1

A 36-year-old female who had been vomiting for 3 days was taken to the emergency department, where the following blood values were measured: **pH = 7.5**; **Pco₂ = 48 mmHg**; and **HCO₃⁻ = 37 mEq/L**.

What acid-base disorder does she have? Does she have a simple or mixed acid-base disorder?



Is it a Simple Metabolic Alkalosis (pH 7.5) or a Mixed Acid-Base Disorder?

Normal $\text{HCO}_3^- = 24\text{mEq/L}$; normal $\text{pCO}_2 = 40\text{mmHg}$

Patient $\text{HCO}_3^- = 37\text{mEq/L}$; patient $\text{pCO}_2 = 48\text{mmHg}$

Above normal **13mEq/L** **8mmHg**

Acid-Base Disturbance	Primary Disturbance	Compensation	Predicted Compensatory Response
Metabolic Acidosis	$\downarrow [\text{HCO}_3^-]$	$\downarrow \text{Pco}_2$	1 mEq/L decrease in $\text{HCO}_3^- \rightarrow 1.2 \text{ mm Hg decrease in } \text{Pco}_2$
Metabolic Alkalosis	$\uparrow [\text{HCO}_3^-]$	$\uparrow \text{Pco}_2$	1 mEq/L increase in $\text{HCO}_3^- \rightarrow 0.6 \text{ mm Hg increase in } \text{Pco}_2$
Respiratory Acidosis			
Acute	$\uparrow \text{Pco}_2$	$\uparrow [\text{HCO}_3^-]$	1 mm Hg increase in $\text{Pco}_2 \rightarrow 0.1 \text{ mEq/L increase in } \text{HCO}_3^-$
Chronic	$\uparrow \text{Pco}_2$	$\uparrow [\text{HCO}_3^-]$	1 mm Hg increase in $\text{Pco}_2 \rightarrow 0.3 \text{ mEq/L increase in } \text{HCO}_3^-$
Respiratory Alkalosis			
Acute	$\downarrow \text{Pco}_2$	$\downarrow [\text{HCO}_3^-]$	1 mm Hg decrease in $\text{Pco}_2 \rightarrow 0.2 \text{ mEq/L decrease in } \text{HCO}_3^-$
Chronic	$\downarrow \text{Pco}_2$	$\downarrow [\text{HCO}_3^-]$	1 mm Hg decrease in $\text{Pco}_2 \rightarrow 0.4 \text{ mEq/L decrease in } \text{HCO}_3^-$

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Please note that the calculation
in Costanzo is faulty! The
numbers here are correct.

Predicted increase in PCO_2 : **13mEq/L x 0.6mmHg/mEq/L** increase in $\text{PCO}_2 = 7.8\text{mmHg}$

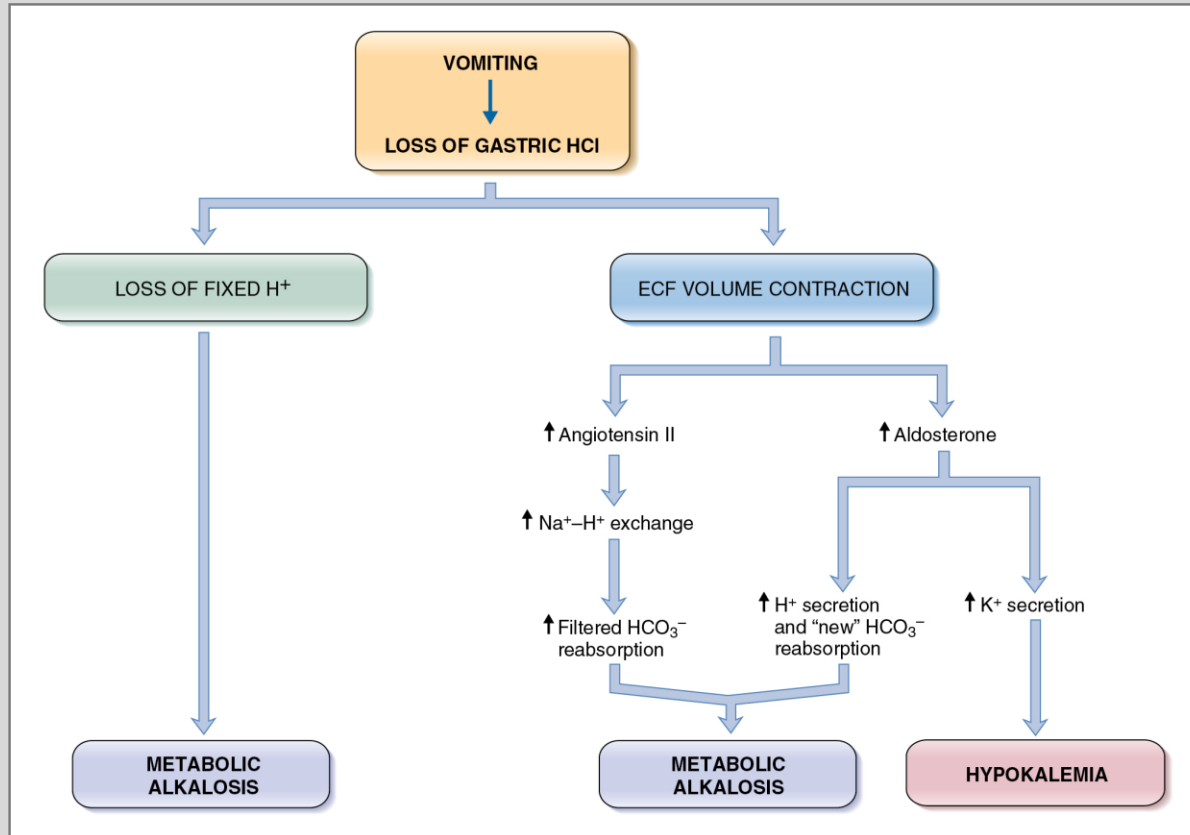
→ Expected $\text{PCO}_2 = 47.8\text{mmHg}$ equals found PCO_2 of 48mmHg

→ Interpretation: expected respiratory compensation for simple metabolic alkalosis

Sequence of Events in Metabolic Alkalosis

1. Loss of fixed H^+ from the stomach through vomiting.
2. Pancreas NOT stimulated to release HCO_3^- which stays in blood and raises pH.
3. Buffering in ECF and ICF. When H^+ leaves cells in exchange for K^+ , hypokalemia results.
4. Respiratory compensation through hypoventilation and increase in PCO_2 and decrease in pH.
5. Ultimate renal correction $\rightarrow$ HCO_3^- excretion until normalization.

Generation and Maintenance of Metabolic Alkalosis with Vomiting



Metabolic Acidosis - Decreased HCO_3^-

Arterial blood profile: **Decreased pH, decreased HCO_3^- and decreased PCO_2**

CAUSES

- **Excessive production or ingestion of fixed H^+**
Diabetic ketoacidosis, lactic acidosis, salicylate poisoning, methanol/formaldehyde poisoning, ethylene glycol poisoning
- **Loss of HCO_3^-**
Diarrhea, Type 2 renal tubular acidosis (type 2 RTA)
- **Inability to excrete fixed H^+**
Chronic kidney disease, Type 1 and type 4 renal tubular acidosis (type 1 RTA, type 4 RTA)

Sample Case #2 – Diabetic Ketoacidosis

A 56-year-old female presents with a 15-year history of type I diabetes mellitus, which has been controlled by careful dietary monitoring and treatment with subcutaneous injections of insulin twice a day. A recent viral illness has resulted in loss of appetite, fever, and vomiting. She becomes short of breath and is admitted to the intensive care unit of the hospital.

Physical examination reveals that the woman is acutely ill. Her mucous membranes are dry, and she has decreased skin turgor. She is breathing deeply and rapidly. A urine sample contains glucose and ketones.

Laboratory tests on her blood yield the following information:

Arterial Blood	Venous Plasma
pH, 7.07	[Na ⁺], 132 mEq/L
Pco ₂ , 18 mm Hg	[Cl ⁻], 94 mEq/L
[HCO ₃ ⁻], 5 mEq/L	[K ⁺], 5.9 mEq/L
	[Glucose], 650 mg/dL

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Please note that the calculation in Costanzo is faulty! The numbers here are correct.

Below normal	19mEq/L	22mmHg
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Predicted decrease in PCO_2 : $19\text{mEq/L} \times 1.2\text{mmHg/mEq/L}$ decrease in $\text{PCO}_2 = 22.8\text{mmHg}$
 → Expected $\text{PCO}_2 = 17.2\text{mmHg}$ equals found PCO_2 of 18mmHg
 → Interpretation: expected respiratory compensation for simple metabolic acidosis

Sample Case #2 – Anion Gap

$$\begin{aligned}\text{Anion gap} &= [\text{Na}^+] - ([\text{Cl}^-] + [\text{HCO}_3^-]) \\ &= 132 - (94 + 5) \\ &= \mathbf{33 \text{ mEq/L}} \quad (8\text{--}16 \text{ mEq/L is normal})\end{aligned}$$

Interpretation: Given her disease and ketones in her urine, **β -OH butyrate** and **acetoacetate** are the likely unmeasured anions

Arterial Blood	Venous Plasma
pH, 7.07	[Na ⁺], 132 mEq/L
Pco ₂ , 18 mm Hg	[Cl ⁻], 94 mEq/L
[HCO ₃ ⁻], 5 mEq/L	[K ⁺], 5.9 mEq/L
	[Glucose], 650 mg/dL

Sequence of Events in Metabolic Acidosis

1. Gain of fixed H^+ through increased production, ingestion, or decreased excretion accompanied by a drop in pH.
2. Excess fixed H^+ is buffered by HCO_3^- in the ECF titrating it away causing a drop in HCO_3^- .
3. Excess fixed H^+ is buffered by organic phosphates and proteins in the ICF (enters with an organic anion). If exchanged for K^+ , hyperkalemia results.
4. Respiratory compensation through hyperventilation (carotid chemoreceptors) and drop in PCO_2 and increase in pH.
5. Ultimate renal correction $\rightarrow H^+$ excreted as ammonium ion (NH_4^+) and increased recovery and production of HCO_3^-

Respiratory Acidosis – Increased CO_2 Through Hypoventilation

Arterial blood profile: **Decreased pH, increased HCO_3^- and increased PCO_2**

CAUSES

- **Inhibition of the medullary respiratory center**
Opiates, barbiturates, anesthetics, lesions of the central nervous system, central sleep apnea, oxygen therapy
- **Disorders of respiratory muscles**
Guillain-Barré syndrome, polio, amyotrophic lateral sclerosis (ALS), multiple sclerosis
- **Airway obstruction**
Aspiration, obstructive sleep apnea, laryngospasm
- **Disorders of gas exchange**
Chronic obstructive pulmonary disease (COPD), pneumonia, pulmonary edema

Sequence of Events in Respiratory Acidosis

1. Retention of CO_2 , increase of PCO_2 , increase of H^+ (and HCO_3^-), drop in pH.
2. Intracellular fluid (ICF) buffering with conversion of CO_2 into H^+ and HCO_3^- inside red blood cells (RBC) and buffering by intracellular proteins and organic phosphates.
3. No respiratory compensation because that is the cause
4. Renal compensation $\rightarrow$ H^+ excreted as ammonium ion (NH_4^+) and increased recovery and production of HCO_3^- ; acute and chronic effects

Respiratory Alkalosis – Excessive Loss of CO_2 Through Hyperventilation

Arterial blood profile: **Increased pH, decreased HCO_3^- and decreased PCO_2**

CAUSES

- **Stimulation of the medullary respiratory center**
Hysterical hyperventilation, Gram-negative septicemia, salicylate poisoning (aspirin), neurologic disorder (tumor, stroke)
- **Hypoxemia**
High altitude, pneumonia, pulmonary embolism
- **Mechanical ventilation**

Sequence of Events in Respiratory Alkalosis

1. Excessive loss of CO_2 , decrease of PCO_2 , decrease of H^+ (and HCO_3^-), increase in pH.
2. Buffering exclusively in intracellular fluid (ICF) of RBCs with CO_2 leaving and intracellular pH going up.
3. No respiratory compensation because that is the cause
4. Renal compensation → Decreased H^+ excretion as ammonium ion (NH_4^+) and decreased recovery and production of HCO_3^- ; acute and chronic effects

Medical Board Style Question

A 24-year-old male is brought to the emergency department after being found unresponsive in his apartment. A rapid clinical workup is initiated, and his basic metabolic panel (BMP) and arterial blood gas (ABG) yield the following values:

Sodium (Na^+): 140 mEq/L; Chloride (Cl^-): 105 mEq/L; Bicarbonate (HCO_3^-): 15 mEq/L; Arterial pH: 7.28; Partial pressure of carbon dioxide (PCO_2): 32 mmHg

Which of the following options correctly calculates his plasma anion gap and correctly identifies his pathophysiological state?

- A. Anion gap of 10 mEq/L, indicating a normal anion gap metabolic acidosis typically caused by severe diarrhea.
- B. Anion gap of 20 mEq/L, indicating a normal anion gap metabolic acidosis secondary to renal tubular acidosis.
- C. Anion gap of 20 mEq/L, indicating a high anion gap metabolic acidosis caused by an accumulation of unmeasured organic anions.
- D. Anion gap of 35 mEq/L, indicating a mixed respiratory acidosis and metabolic alkalosis.
- E. Anion gap of 5 mEq/L, indicating a profound hypoalbuminemia with no true primary acid-base disturbance.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

Lecture Feedback Form

<https://www.surveymonkey.com/r/NYITCOMLectureFeedback>